

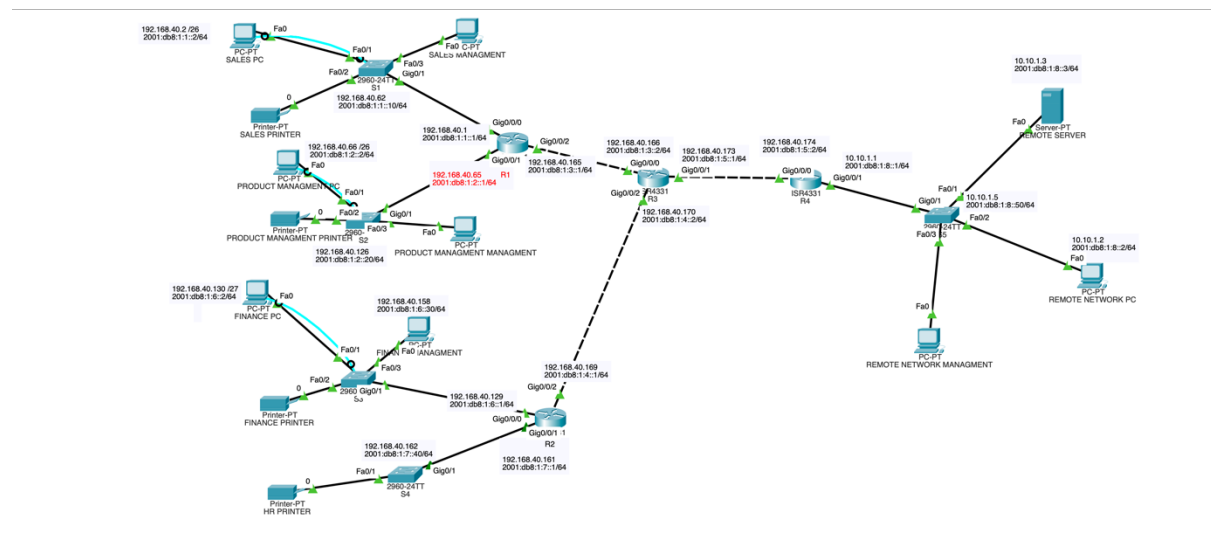
Enterprise Network Design

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Scenario

Your company, currently located at the Eastern part of Thessaloniki plans to build a new building to support non-technical divisions at the center of the city. As a network administrator, you have been asked to design and implement a network to support operations for the following departments:

- - Sales: 50 PCs, 2 Printers, 0 laptops
- - Product Management: 55 PCs, 3 Printers, 0 laptops
- - Finance: 23 PCs, 2 printers, 0 laptops
- - HR: 0 PCs, 1 printer
- - Remote Network: 1 Server, 1 PC



Hardware

Routers: I used four 4331 routers:

- a. R1 for the **Sales** and the **Product Management Network**
- b. R2 for the **Finance** and the **HR Network**
- c. R3 as intermediary between R1 – R2 – R3
- d. R4 for the **Remote Network**

Switches: I used five **2960** switches:

- e. S1 for the **Sales Network**
- f. S2 for the **Product Management Network**
- g. S3 for the **Finance Network**
- h. S4 for the **HR Network**

IP Addressing

Network	IPv4 Subnet	Subnet Mask	Usable IPs	Default Gateway	IPv6 Subnet
SALES - VLAN 10	192.168.40.0/26	255.255.255.192	192.168.40.1 – 192.168.40.62	192.168.40.1	2001:db8:1:1::/64
PR. MANAGEMENT - VLAN 20	192.168.40.64/26	255.255.255.192	192.168.40.65 – 192.168.40.126	192.168.40.65	2001:db8:1:2::/64
FINANCE - VLAN 30	192.168.40.128/27	255.255.255.224	192.168.40.129 – 192.168.40.158	192.168.40.129	2001:db8:1:6::/64
HR - VLAN 40	192.168.40.160/30	255.255.255.252	192.168.40.161–192.168.40.162	192.168.40.161	2001:db8:1:7::/64
R1 - R3	192.168.40.164/30	255.255.255.252	192.168.40.165 – 192.168.40.166	192.168.40.165	2001:db8:1:3::/64
R2 - R3	192.168.40.168/30	255.255.255.252	192.168.40.169 – 192.168.40.170	192.168.40.169	2001:db8:1:4::/64
R3 - R4	192.168.40.172/30	255.255.255.252	192.168.40.173 – 192.168.40.174	192.168.40.173	2001:db8:1:5::/64
REMOTE NETWORK	10.10.1.0/29	255.255.255.248	10.10.1.1-10.10.1.6	10.10.1.1	2001:db8:1:8::/64

Device	Interface	IPv4 Address	IPv4 Subnet Mask	Default Gateway	IPv6 Addresses/Prefix	Link Local Address	IPv6 Default Gateway
R1	Gig0/0/0.10	192.168.40.1	255.255.255.192	N/A	2001:db8:1:1::1/64	fe80::1	N/A
	Gig0/0/0.1.20	192.168.40.65	255.255.255.192	N/A	2001:db8:1:2::1/64	fe80::1	N/A
	Gig0/0/0.2	192.168.40.165	255.255.255.252	N/A	2001:db8:1:3::1/64	fe80::1	N/A
R2	Gig0/0/0.30	192.168.40.129	255.255.255.224	N/A	2001:db8:1:6::1/64	fe80::1	N/A
	Gig0/0/0.1.40	192.168.40.161	255.255.255.252	N/A	2001:db8:1:7::1/64	fe80::1	N/A
	Gig0/0/0.2	192.168.40.169	255.255.255.252	N/A	2001:db8:1:4::1/64	fe80::1	N/A
R3	Gig0/0/0	192.168.40.166	255.255.255.252	N/A	2001:db8:1:3::2/64	fe80::2	N/A
	Gig0/0/0.1	192.168.40.173	255.255.255.252	N/A	2001:db8:1:5::1/64	fe80::2	N/A
	Gig0/0/0.2	192.168.40.170	255.255.255.252	N/A	2001:db8:1:4::2/64	fe80::2	N/A
R4	Gig0/0/0	192.168.40.174	255.255.255.252	N/A	2001:db8:1:5::2/64	fe80::1	N/A
	Gig0/0/0.1	10.10.1.1	255.255.255.248	N/A	2001:db8:1:8::1/64	fe80::1	N/A
S1	VLAN 99	192.168.40.62	255.255.255.192	192.168.40.1	2001:db8:1:1::10/64	fe80::3	2001:db8:1:1::1
S2	VALN 99	192.168.40.126	255.255.255.192	192.168.40.65	2001:db8:1:2::20/64	fe80::3	2001:db8:1:2::1
S3	VLAN 99	192.168.40.158	255.255.255.224	192.168.40.129	2001:db8:1:6::30/64	fe80::3	2001:db8:1:6::1
S4	VLAN 99	192.168.40.162	255.255.255.252	192.168.40.161	2001:db8:1:7::40/64	fe80::3	2001:db8:1:7::1
S5	VLAN 99	10.10.1.5	255.255.255.248	10.10.1.1	2001:db8:1:8::50/64	fe80::4	2001:db8:1:8::1
SALES PC	Fa0	192.168.40.2	255.255.255.192	192.168.40.1	2001:db8:1:1::2/64	fe80::2	2001:db8:1:1::1
SALES PRINTER	0	N/A	N/A	192.168.40.1	N/A	N/A	2001:db8:1:1::1
PRODUCT MANAGEMENT PC	Fa0	192.168.40.66	255.255.255.192	192.168.40.65	2001:db8:1:2::2/64	fe80::2	2001:db8:1:2::1
PRODUCT MANAGEMENT PRINTER	0	N/A	N/A	192.168.40.65	N/A	N/A	2001:db8:1:2::1
FINANCE PC	Fa0	192.168.40.130	255.255.255.224	192.168.40.129	2001:db8:1:6::2/64	fe80::2	2001:db8:1:6::1
FINANCE PRINTER	0	N/A	N/A	192.168.40.129	N/A	N/A	2001:db8:1:6::1
HR PRINTER	0	N/A	N/A	192.168.40.161	N/A	N/A	2001:db8:1:7::1
REMOTE NETWORK PC	Fa0	10.10.1.2	255.255.255.248	10.10.1.1	2001:db8:1:8::2/64	fe80::2	2001:db8:1:8::1
REMOTE NETWORK SERVER	Fa0	10.10.1.3	255.255.255.248	10.10.1.1	2001:db8:1:8::3/64	fe80::3	2001:db8:1:8::1

IPv4, IPv6 Assignments:

End devices

Routers IPv4, IPv6

```
R1(config-if)#ex
R1(config)#int gi0/0/1.20
R1(config-subif)#ipv6 ad
R1(config-subif)#ipv6 address 2001:db8:1:2::1/64
R1(config-subif)#ipv6 ad
R1(config-subif)#ipv6 address fe80::1
% Incomplete command
```

```
Router(config)#hostname R1
R1(config)#int gi0/0/0
R1(config-if)#ip adr
R1(config-if)#ip ad
R1(config-if)#ip address 192.168.40.1 255.255.255.192
R1(config-if)#no shutdown
```

```
R1(config-subif)#en
R1(config-subif)#encapsulation q
R1(config-subif)#encapsulation do
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip ad
R1(config-subif)#ip address 192.168.40.65 255.255.255.192
R1(config-subif)#ex
```

Security

Login Password (line con 0):

```
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#line con 0
S1(config-line)#pas
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#
```

Privileged EXEC Mode Secret Password & Service Password Encryption

Only Authorized Users!

```
S1>en
S1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#pas
S1(config)#ser
S1(config)#service pas
S1(config)#service password-encryption ena
S1(config)#service password-encryption en
S1(config)#service password-encryption
S1(config)#ex
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy run
S1#copy running-config st
S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#pas
S1(config)#en
S1(config)#enable s
S1(config)#enable secret class
S1(config)#
```

Message of the Day Banner

```
% Incomplete command.
S1(config)#hostname S1
S1(config)#banner motd "Only Authorized Users!"
S1(config)#
```

SSH for Routers

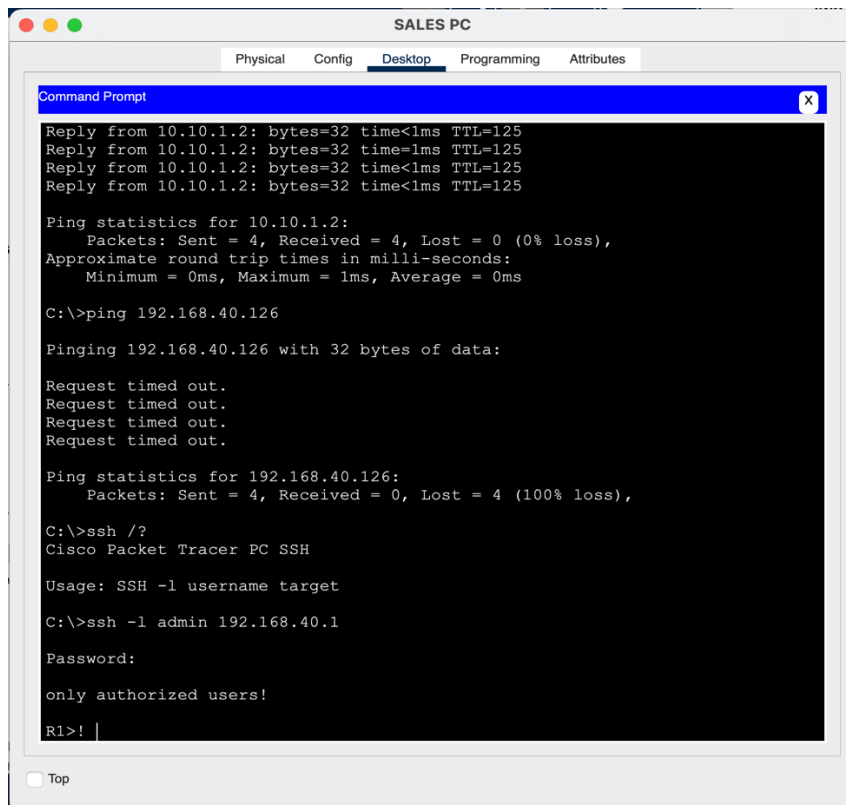
```

R1#sec
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#sec
R1(config)#security pas
R1(config)#security password min
R1(config)#security password min-length 5
R1(config)#no ip
R1(config)#no ip domain-lookup
R1(config)#ip domain-name networks.com
R1(config)#username admin secret admin123
R1(config)#crypto key generate rsa
The name for the keys will be: R1.networks.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#login block-for 180 attempts 4 within 120
*Mar 1 2:16:9.685: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 15
R1(config-line)#tran
R1(config-line)#transport in
R1(config-line)#transport input s
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exec
R1(config-line)#exec-timeout 6
R1(config-line)#

```



SSH for Switches

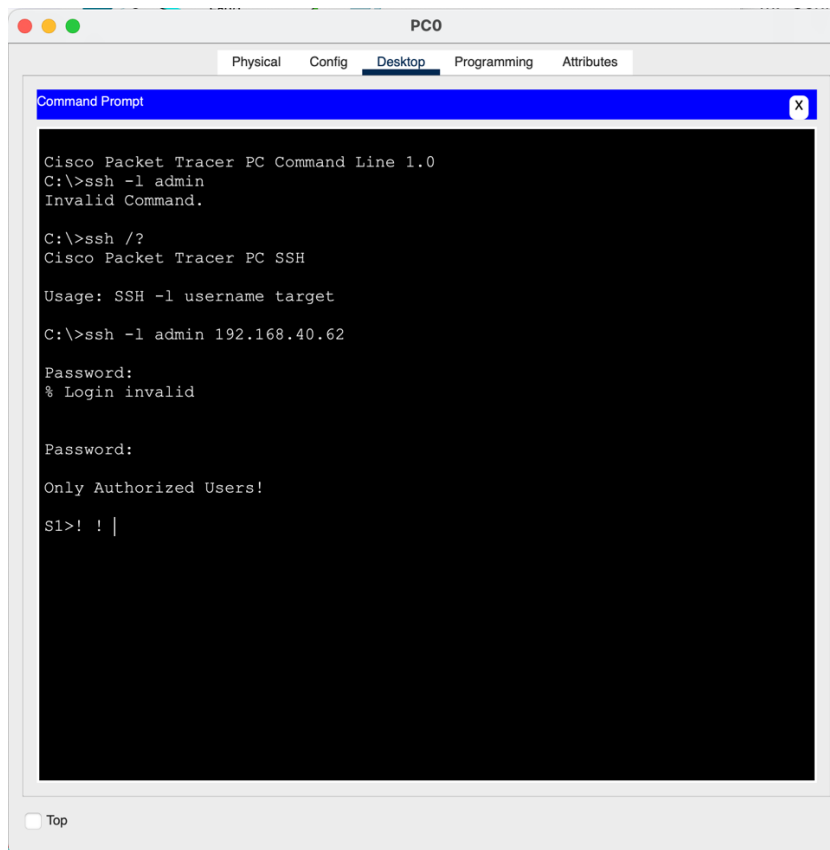
```

[OK]
S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#no ip
S2(config)#no ip dom
S2(config)#no ip domain
S2(config)#no ip domain-lookup
S2(config)#ip domain
S2(config)#ip domain-name networks.com
S2(config)#username admin secret admin123
S2(config)#crypto key generate rsa
The name for the keys will be: S2.networks.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

S2(config)#line vty 0 15
*Mar 1 2:47:28.9: %SSH-5-ENABLED: SSH 1.99 has been enabled
S2(config-line)#trasn
S2(config-line)#transport input ssh
S2(config-line)#login local
S2(config-line)#exe
S2(config-line)#exec-timeout
% Incomplete command.
S2(config-line)#ex
S2(config-line)#exec
S2(config-line)#exec-timeout 6
S2(config-line)#

```



- All Routers can be accessed through SSH.
- Switches can be access via SSH if there is a "Management" pc in the network which I added solely for this reason, and it is not configured to communicate effectively within the network

VLANs

VLAN CREATION

```
S3(config)#vlan 30
S3(config-vlan)#name FINANCE
S3(config-vlan)#end
S3#
```

- All VLANs were created in every switch in the project
- Switches' Ports were assigned properly to their respected VLAN

```
S1(config)#int fa0/2
S1(config-if)#swi
S1(config-if)#switchport m
S1(config-if)#switchport mode ac
S1(config-if)#switchport mode access
S1(config-if)#sw
S1(config-if)#switchport ac
S1(config-if)#switchport access vlan 10
S1(config-if)#
```

1	default	active	Fa0/1, Fa0/2, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12,
	Fa0/13		Fa0/14, Fa0/15, Fa0/16,
	Fa0/17		Fa0/18, Fa0/19, Fa0/20,
	Fa0/21		Fa0/22, Fa0/23, Fa0/24,
	Gig0/1		Gig0/2
99	Managment	active	Fa0/3
100	Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Management Interface VLAN 99 Creation and IP Assignment

```

S1(config)#int vlan 99
S1(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up

S1(config-if)#ip ad
S1(config-if)#ip address 192.168.40.62 255.255.255.192
S1(config-if)#ip de
S1(config-if)#ip default-gat
S1(config-if)#ip default-gateway 192.168.40.1
S1(config-if)#ex

Vlan1          unassigned      YES manual administratively down down
Vlan99         192.168.40.62    YES manual up                up

```

Native VLAN Assignment

```

Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#int gi0/1
S1(config-if)#sw
S1(config-if)#switchport m
S1(config-if)#switchport mode t
S1(config-if)#switchport mode trunk
S1(config-if)#sw
S1(config-if)#switchport non
S1(config-if)#switchport nonegotiate
S1(config-if)#sw
S1(config-if)#switchport tr
S1(config-if)#switchport trunk ?
    allowed  Set allowed VLAN characteristics when interface is in trunking
mode
    native   Set trunking native characteristics when interface is in trunking
mode
S1(config-if)#switchport trunk na
S1(config-if)#switchport trunk native vlan 100
S1(config-if)#

```

Trunk Configuration

```

S1(config)#int gi0/1
S1(config-if)#sw
S1(config-if)#switchport mo
S1(config-if)#switchport mode t
S1(config-if)#switchport mode trunk

S1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up

S1(config-if)#sw
S1(config-if)#switchport tr
S1(config-if)#switchport trunk allo
S1(config-if)#switchport trunk allowed vlan 10, 20, 30, 40, 99, 100
^
% Invalid input detected at '^' marker.

S1(config-if)#ss
S1(config-if)#sw
S1(config-if)#switchport tr
S1(config-if)#switchport trunk allow
S1(config-if)#switchport trunk allowed vlan 10,20,30,40,99,100
S1(config-if)#

```

Routing

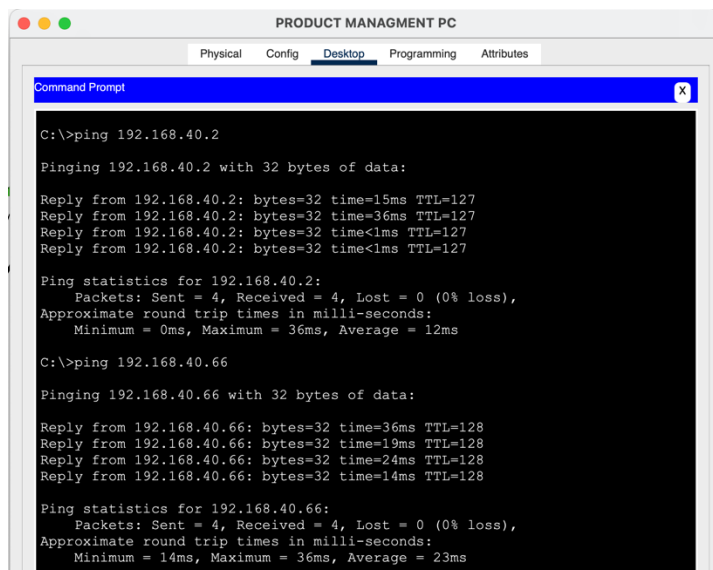
IPv4 Routing

Enter configuration commands, one per line. End with CNTL/Z.

```

R4(config)#ip route 192.168.40.0 255.255.255.192 192.168.40.173
R4(config)#ip route 192.168.40.64 255.255.255.192 192.168.40.173
R4(config)#ip route 192.168.40.128 255.255.255.224 192.168.40.173
R4(config)#ip route 192.168.40.160 255.255.255.252 192.168.40.173
R4(config)#

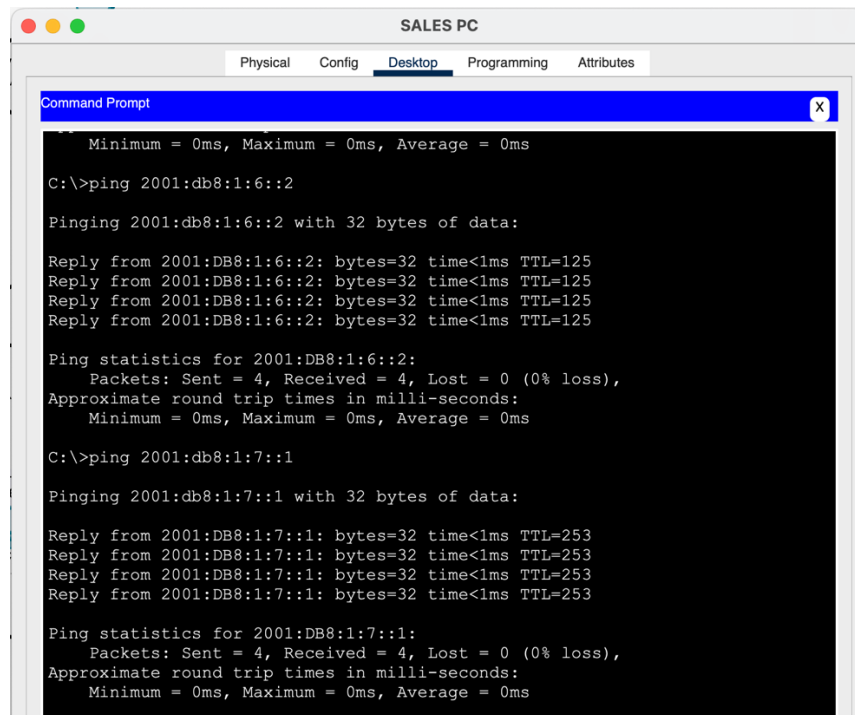
```



Routing

IPv6 Routing

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ipv6 route 2001:db8:1:6::/64 2001:db8:1:3::2
R1(config)#ipv6 route 2001:db8:1:7::/64 2001:db8:1:3::2
R1(config)#ipv6 route 2001:db8:1:8::/64 2001:db8:1:3::2
R1(config)#
```



PASSWORDS

- **Login:** cisco (line con 0)
- **EXEC Mode:** class (enable)
- **SSH:**
 - **Username:** admin
 - **Password:** admin123